25901094

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Value Addition Course (VAC) - MG4VACBCH200 - NARCOTICS AND PSYCHOTROPIC SUBSTANCES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice questions

Answer all questions

Each question carries **1** marks

1. Metabolism of drugs refers to

option c

c). Chemical alteration of the drug

2. Blockade of 5-HT2A receptors by antipsychotic agents helps in reducing

Option c -negative symptoms

3. Addiction to———often leads to legal issues, financial problems and social isolation

Option b -Narcotics

4. Policy-making in narcotics study helps to:

Option d- Balance medical necessity and misuse

5. The main purpose of the NDPS Act is to:

Option b-control misuse and regulate medical use of narcotics

[1x5 = 5]

Part B

Fill in the Blanks questions

Answer all questions Each question carries 2 marks

6. Narcotic drugs primarily act as___pain relievers.

7. Barbiturates increase the duration of opening of Chloridechannels.

8. MAOIs increase monoamine levels by inhibiting monoamine oxidase.

9. A drug that activates a receptor is called an _agonist_.

10. In classicalconditioning, environmental cues associated with drug use become triggers for craving.

[2x5 = 10]

Part C

Short Answer Type Questions

Answer any 5 questions Each question carries 3 marks

11.What is pharmacokinetics?

Pharmacokinetics is the study of how an organism affects the drug. It involves four key stages: absorption, distribution, metabolism, and excretion.(1 mark)

Explain each stage in one or two sentences (one mark each for two stages)

12.Define anxiolytics and mention two clinical uses.

Anxiolytics are drugs that reduce anxiety, fear, tension, and excessive worry. (1 mark) They are used in conditions like Generalized Anxiety DisorderGAD, panic disorder, social anxiety, and phobias. (any two, 2 marks)

13.Explain the uses of oxycodone.

Oxycodone is primarily used for the *management of moderate to severe pain*, especially when other pain relievers are ineffective. It is commonly prescribed for post-operative pain, cancer-related pain, and chronic pain conditions such as osteoarthritis or back injuries. (any three use, I mark each)

14.Differentiate between pharmacological treatment and behavioural therapy in addiction management.

Needed only brief description of the following.

- **Medical/Pharmacological Strategies**

The purpose of detoxification is the safe withdrawal from the drug under medical supervision. Usually, the adopted methods are gradual tapering of the narcotic dose by using substitution medications (e.g. methadone) and symptomatic treatment for withdrawal (e.g., clonidine for anxiety, pain relievers)

- **Psychological and behavioural therapies**

Includes- behavioural therapies, Cognitive-behavioural therapy (CBT) and motivational interviewing.

Cognitive Behavioural Therapy (CBT)- CBT helps to identify and change negative thoughts and behaviours related to drug use by counselling (Individual/group counselling) to address the underlying issues and promote recovery.

Motivational Interviewing (MI)- Strengthens personal motivation and commitment to recovery.

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15.Describe the psychological effects produced by LSD.

- Bad trips (anxiety, panic)
- May trigger psychosis in susceptible individuals
- HPPD (persistent visual disturbances)

16.Explain how addiction affects decision-making centers of the brain.

Only short description is needed.

17. Explain the significance of withdrawal symptoms in diagnosis of dependence.

Withdrawal Symptoms

Withdrawal occurs when drug use is abruptly reduced or stopped after dependence has developed. It is caused by the brain's inability to immediately readjust after prolonged neurochemical alterations.

General Features:

- Symptoms are typically opposite to the drug's acute effects.
- Withdrawal reinforces continued drug use through negative reinforcement, where taking the drug alleviates withdrawal discomfort.

[3x5 = 15]

Part D

Short Essay Type Questions

Answer any **4** questions

Each question carries **5** marks

18.Explain the role of MDMA in increasing oxytocin and serotonin levels and how this leads to its empathogenic and entactogenic effects.

- MDMA (3,4-methylenedioxymethamphetamine) is a synthetic psychoactive drug with both stimulant and hallucinogenic properties.

Mechanism of Action

- Main effect: Massive release of serotonin (5-HT) and inhibition of its reuptake → → Mood elevation→ Emotional bonding → Enhanced sensory perception
- Also increases dopamine and norepinephrine → → Increased energy, alertness
- Stimulates oxytocin release, contributing to prosocial and empathogenic effects.

19.Explain how the brain's reward system contributes to the misuse of narcotic drugs.

All addictive drugs stimulate the brain'smesolimbic dopamine pathway**,** also called thereward circuit**.**

This pathway runs from theventral tegmental area (VTA)to thenucleus accumbens (NAc)and connects with theamygdala, hippocampus, and prefrontal cortex**.**

Normally, this system releases moderate dopamine for natural rewards like food and social bonding.

Addictive drugs, however, cause alarge, unnatural surge of dopamine, creating intense pleasure and reinforcing drug use.

With repeated use, the brain learns to link the drug with reward, strengthening these pathways and making drug-seeking compulsive.

Over time, natural rewards lose importance, and the drug becomes the main source of motivation.

20.Explain how awareness programmes support families and communities in preventing substance abuse.

Importance of Awareness in Prevention

- ✓ Awareness programs are important for preventing substance abuse.
- ✓ They teach people about the dangers and long-term effects of drugs and alcohol.
- ✓ Many individuals, especially young people, start using substances because of ignorance, peer pressure, or curiosity.
- ✓ These programs provide correct information about the physical, mental, and social harms of addiction and help people make informed choices.
- ✓ Awareness programs also clear up myths, reduce stigma, and encourage healthy lifestyles. Preventing substance abuse is more effective than treating it.

Targeting Vulnerable Populations

- ✓ Young adults, adolescents, and marginalized communities are particularly vulnerable to substance abuse.
- ✓ Awareness programmes tailored to these groups can address specific risk factors such as unemployment, academic stress, family issues, and lack of recreational activities.
- ✓ School and college-based campaigns, community outreach, and media-driven initiatives can reach these populations early, fostering resilience and promoting coping strategies.
- ✓ When people understand the risks and recognize early warning signs, they are more likely to seek help or avoid harmful substances altogether.

Reducing Health Burden and Social Costs

- ✓ Substance abuse leads to a cascade of health problems, including chronic diseases, mental illness, and increased risk of accidents and violence.
- ✓ It also places a heavy burden on healthcare systems and law enforcement.
- ✓ Awareness programmes can reduce these costs by encouraging early intervention and promoting rehabilitation.
- ✓ They also help families and communities understand how to support affected individuals, reducing isolation and improving recovery outcomes.
- ✓ In the long run, prevention through education is more cost-effective than managing the consequences of addiction.

21.Explain natural narcotics and their medical implications.

Natural Narcotics

These are directly derived from the opium poppy plant (*Papaver somniferum*). The primary natural narcotics include *morphine*, *codeine*, and *thebaine*. Morphine is the most potent and widely used for severe pain relief, especially in post-surgical and cancer patients. Codeine is

milder and often used in cough syrups and mild painkillers. These substances act on opioid receptors in the brain to reduce pain perception and induce sedation.

Morphine is a powerful opioid analgesic used to treat moderate to severe pain, but its strong effects on the brain's reward system make it highly addictive and prone to misuse.

Morphine is widely used in clinical settings for *acute and chronic pain management*. It is especially valuable in treating pain associated with surgery, trauma, cancer, and palliative care. It can be administered orally, intravenously, subcutaneously, or via epidural routes depending on the clinical need. In emergency medicine, morphine is also used to relieve pain from myocardial infarction (heart attack) and to reduce anxiety

22.Explain social stigma and discrimination faced by substance abusers.

- ✓ The social repercussions of substance abuse are profound and far-reaching.
- ✓ It often leads to strained relationships with family, friends, and colleagues due to behavioural changes, dishonesty, and neglect of responsibilities.
- ✓ Substance abusers may isolate themselves or become aggressive, causing emotional distress to those around them.
- ✓ In communities, drug abuse can contribute to increased crime rates, including theft, violence, and drug trafficking.
- ✓ Educational and professional achievements are frequently disrupted, as individuals may drop out of school or lose their jobs.
- ✓ Social stigma and discrimination further marginalize users, making reintegration into society difficult.
- ✓ These consequences not only affect the individual but also destabilize families and communities.

23.Describe the use of narcotic drugs in the treatment of diarrhoea.

Narcotics like loperamide and diphenoxylate are used to manage severe diarrhoea. These drugs

slow intestinal motility by acting on opioid receptors in the gastrointestinal tract, allowing more time for water absorption and reducing stool frequency.

[5x4 = 20]